

Preface

Topology

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Chapter 1

Topological Spaces

§ 1.1. Topological Spaces

Definition 1.1.1. A topology on a set X is a collection T of subsets of X such that:

- (1) $\emptyset, X \in T$;
- (2) if $\{U_i\}_{i \in I} \subseteq T$, then $\bigcup_{i \in I} U_i \in T$;
- (3) if $\{U_i\}_{i=1}^n \subseteq T$, then $\bigcap_{i=1}^n U_i \in T$.

We call the pair (X, T) a topological space. The subsets of X contained in T are called open sets.

Example 1.1.1. Let $X = \{a, b, c\}$. Then $T_1 := \{\emptyset, \{a, b, c\}, \{b\}, \{a, b\}, \{b, c\}\}$ and $T_2 := \{\emptyset, \{b\}, \{a, b, c\}\}$ are topologies on X . However, $T_3 = \{\emptyset, \{a\}, \{b\}, \{a, b, c\}\}$ is *not* a topology on X . Note that $\{a\} \cup \{b\} = \{a, b\}$, which is not in T_3 .

Example 1.1.2 (Every set admits a topology). Let X be a set. The *discrete topology* on X is $\mathcal{P}(X)$. The *trivial topology* on X is $\{\emptyset, X\}$.

Definition 1.1.2. Let T_1 and T_2 be topologies on a set X . If $T_1 \subseteq T_2$, then we say T_2 is finer than T_1 , and likewise that T_1 is coarser than T_2 .

Example 1.1.3. Let $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$ be vectors in $\mathbb{R}^n$. Define $d(\vec{x}, \vec{y}) := \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$. This is a metric on $\mathbb{R}^n$, and we will see this induces a topology on $\mathbb{R}^n$.

Definition 1.1.3. Let (X, d) be a metric space, $x \in X$, and $\epsilon > 0$. The open ball centered at x with radius ϵ is denoted $B(x, \epsilon) = \{y \in X \mid d(x, y) < \epsilon\}$.

Definition 1.1.4. Let (X, d) be a metric space. A set $U \subseteq X$ is *open* if for all $x \in U$, there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$. The collection of open sets of X is denoted τ_X .

Proposition 1.1.1. Let (X, d) be a metric space. Then (X, τ_X) is a topological space.

Proof. Clearly $\emptyset, X \in \tau_X$. Let $\{U_i\}_{i \in I} \subseteq \tau_X$ be an arbitrary family of open sets. Let $x \in \bigcup_{i \in I} U_i$. Then $x \in U_i$ for some i . So there exists an $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U_i \subseteq \bigcup_{i \in I} U_i$. Now let $\{U_i\}_{i=1}^n$ be a finite collection of open sets. Let $x \in \bigcap_{i=1}^n U_i$. Then $x \in U_i$ for all $i = 1, \dots, n$. So there exists $\epsilon_i > 0$ such that $B(x, \epsilon_i) \subseteq U_i$ for all $i = 1, \dots, n$. Define $\epsilon := \min\{\epsilon_1, \dots, \epsilon_n\}$. Then $B(x, \epsilon) \subseteq U_i$ for all $i = 1, \dots, n$. Hence $B(x, \epsilon) \subseteq \bigcap_{i=1}^n U_i$. Thus (X, τ_X) is a topological space. $\square$

Example 1.1.4. Using the same terminology from Example 1.1.3, since $d(\vec{x}, \vec{y})$ is a metric, $(\mathbb{R}^n, \tau_{\mathbb{R}^n})$ is a topological space.

§ 1.2. Topologies Generated by a Basis

Definition 1.2.1. Let X be a set. A basis for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

- (1) for all $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$.
- (2) If $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Remark. Let (X, d) be a metric space and $\mathcal{B}$ a basis for a topology on X . It follows from the above definition that for any $U \in \tau_X$, there exists an open basis element $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Example 1.2.1. The sets $\mathcal{B}_1 := \{(a, b) \mid a < b\}$ and $\mathcal{B}_2 := \{(a, b) \mid a < b, a, b \in \mathbb{Q}\}$ are bases for a topology on $\mathbb{R}$. These definitions extend nicely to $\mathbb{R}^n$ by considering open balls.

Example 1.2.2 (Every set admits a basis for a topology). Let X be a set. Then $\mathcal{B} := \{\{p\} \mid p \in X\}$ is a basis for a topology on X .

Proposition 1.2.1. Let X be a set and $\mathcal{B}$ a basis for a topology on X . Let τ be the set containing the empty set and every subset of X which is equal to a union of basis elements of $\mathcal{B}$. Then (X, τ) is a topological space.

Proof.

□

Definition 1.2.2. Let X be a set and $\mathcal{B}$ a topology on X . The topology defined in Proposition 1.2.1 is called the topology generated by $\mathcal{B}$.

Example 1.2.3. From Example 1.2.1, $\mathcal{B}_1$ and $\mathcal{B}_2$ generate the standard topology on $\mathbb{R}^n$. From Example 1.2.2, $\mathcal{B}$ generates the discrete topology on X .

§ 1.3. Product Topologies

Definition 1.3.1. Let (X, τ_1) and (Y, τ_2) be topological spaces. The product topology on $X \times Y$ is the topology generated by the basis $\mathcal{B} := \{U \times V \mid U \in \tau_1, V \in \tau_2\}$.

Remark. Note that $\mathcal{B}$ is not necessarily a topology on $X \times Y$. Indeed, consider $X = Y = \{0, 1\}$ in the discrete topology. Then the sets $\{0\} \times \{0\}$ and $\{0\} \times \{1\}$ are in $\mathcal{B}$ as per the above definition, but their union $\{(0, 0), (0, 1)\}$ is not. Hence $\mathcal{B}$ is not a topology.

Example 1.3.1. Recall from Example 1.2.1 that:

$$\mathcal{B} := \{(a, b) \mid a < b\}$$

generates the standard topology on $\mathbb{R}$. Consider the following to bases for a topology on $\mathbb{R}^2$:

$$\mathcal{B}_1 := \{B(x, r) \mid x \in \mathbb{R}^2, r > 0\},$$

$$\mathcal{B}_2 := \{I_1 \times I_2 \mid I_1, I_2 \in \mathcal{B}\}.$$

The basis elements of $\mathcal{B}_1$ are open balls of radius r , whilst the basis elements of $\mathcal{B}_2$ are open squares. Note that the basis $\mathcal{B}_2$ interprets $\mathbb{R}^2$ as the product $\mathbb{R} \times \mathbb{R}$.

This idea can be further generalized to $\mathbb{R}^3$. Consider:

$$\mathcal{B}_3 := \{B(x, r) \mid x \in \mathbb{R}^3, r > 0\}$$

$$\mathcal{B}_4 := \{I_1 \times I_2 \times I_3 \mid I_1, I_2, I_3 \in \mathcal{B}\}$$

$$\mathcal{B}_5 := \{B \times I \mid B \in \mathcal{B}_1, I \in \mathcal{B}\}.$$

The basis $\mathcal{B}_4$ interprets $\mathbb{R}^3$ as $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$, whilst the basis $\mathcal{B}_5$ interprets $\mathbb{R}^3$ as $\mathbb{R}^2 \times \mathbb{R}$.

Proposition 1.3.1. *The bases $\mathcal{B}_1$ and $\mathcal{B}_2$ from Example 1.3.1 generate the same topologies on $\mathbb{R}^2$.*

Proof.

□

§ 1.4. Subspace Topology

Proposition 1.4.1. *Let $(X, \mathcal{T})$ be a topological space. If $Y \subseteq X$, then $\mathcal{T}_Y := \{Y \cap U \mid U \in \mathcal{T}_X\}$ is a topology on Y .*

Proof.

□

Definition 1.4.1. Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$. The *subspace topology* on Y is the topology defined in Proposition 1.4.1. We call Y a subspace of X if it is paired with the subspace topology.

Example 1.4.1. Consider the interval $[2, 4)$ with the subspace topology inherited from $\mathbb{R}$. Note that $(3, 4)$ is open in $[2, 4)$ and $\mathbb{R}$, while $[2, 3)$ is open in $[2, 4)$ but *not* open in $\mathbb{R}$.

Example 1.4.2. Consider the integers $\mathbb{Z}$ with the subspace topology inherited from $\mathbb{R}$. For any $n \in \mathbb{Z}$, we have $\{n\} = \mathbb{Z} \cap (n - \frac{1}{2}, n + \frac{1}{2})$. Since $(n - \frac{1}{2}, n + \frac{1}{2})$ is open in $\mathbb{R}$, each singleton in $\mathbb{Z}$ is open. Hence the subspace topology on $\mathbb{Z}$ inherited from $\mathbb{R}$ is the discrete topology.

Proposition 1.4.2. *If $\mathcal{B}$ is a basis for a topology on a set X , and Y is a subset of X , then $\mathcal{B}_Y := \{Y \cap U \mid U \in \mathcal{B}\}$ is a basis for Y .*

Proof.

□

Example 1.4.3. Recall that $\mathcal{B} := \{(a, b) \mid a < b\}$ is a basis for the standard topology on $\mathbb{R}$. Consider $\mathcal{B}_{\mathbb{Q}} := \{(a, b) \cap \mathbb{Q} \mid a < b\}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we have that $(a, b) \cap \mathbb{Q} \neq \emptyset$. In fact, each nonempty open set of $\mathbb{Q}$ with this particular subspace topology is infinite, hence it is not the discrete topology.

§ 1.5. Closed Sets

Definition 1.5.1. A subset A of a topological space X is closed in X if A^c is open.

Example 1.5.1. Let (X, T) be a topological space. Then $\emptyset$ and X are closed.

Example 1.5.2. If $x \in \mathbb{R}^n$, then $\{x\}$ is closed in $\mathbb{R}^n$. Indeed, for any $p \in \mathbb{R}^n \setminus \{x\}$, we have that $p \in B\left(p, \frac{\|p-x\|}{3}\right) \subseteq \mathbb{R}^n \setminus \{x\}$. Since $\mathbb{R}^n \setminus \{x\}$ is open, $\{x\}$ must be closed.

Example 1.5.3. Given any $x \in \mathbb{R}^n$, the set $\{y \in \mathbb{R}^n \mid \|x - y\| \leq \epsilon\}$ is closed. The proof of this fact follows identically to Example 1.5.2.

Example 1.5.4. Consider the topological space $(X, \mathcal{P}(X))$. Since every set is open, every set must be closed.

Proposition 1.5.1. *Let X be a topological space. Then:*

- (1) $\emptyset$ and X are closed;
- (2) If $\{C_i\}_{i \in I}$ is an arbitrary family of closed sets, then $\bigcap_{i \in I} C_i$ is closed;
- (3) If $\{C_i\}_{i=1}^n$ is a finite collection of closed sets, then $\bigcup_{i=1}^n C_i$ is closed.

Proof. This follows from De Morgan's laws and Definition 1.1.1.

□

Example 1.5.5. Every finite subset of $\mathbb{R}^n$ is closed. Indeed, any finite subset can be written as the finite union of singletons, which are closed in $\mathbb{R}^n$ by Example 1.5.2.

Proposition 1.5.2. *Let Y be a subspace of a topological space X . Then a set $A \subseteq Y$ is closed in Y if and only if $A = Y \cap C$, where C is a closed set in X .*

Proof.

□

§ 1.6. Closure and Interior

Definition 1.6.1. Let (X, T) be a topological space and $A \subseteq X$.

- (1) The interior of A is $A^\circ := \bigcup \{U \mid U \subseteq A, U \text{ open}\}$.
- (2) The closure of A is $\bar{A} := \bigcap \{C \mid C \supseteq A, C \text{ closed}\}$.

Remark. Clearly $A^\circ \subseteq A \subseteq \bar{A}$. The interior of any set is open and is the largest open subset containing it. The closure of any set is closed and is the smallest closed subset contained in it. A set is open if and only if it is equal to its interior. A set is closed if and only if it is equal to its closure. If C is any closed set in X and $A \subseteq C$, then $\bar{A} \subseteq C$.

Example 1.6.1. If $F \subseteq \mathbb{R}^n$ is finite then $F^\circ = \emptyset$ and $\bar{F} = F$.

Example 1.6.2. Consider the interval $(2, 3]$ in $\mathbb{R}$. Then $(2, 3]^\circ = (2, 3)$ and $\overline{(2, 3]} = [2, 3]$.

Example 1.6.3. In the discrete topology, every set is open and closed, hence every set equals its interior and closure.

Example 1.6.4. Consider the set $A := \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ as a subset of $\mathbb{R}$. Then $A^\circ = \emptyset$ and $\bar{A} = A \cup \{0\}$.

Proposition 1.6.1. Let (X, T) be a topological space and $A \subseteq X$.

- (1) $p \in \bar{A}$ if and only if for all open sets U in X containing p , we have $U \cap A \neq \emptyset$.
- (2) If $\mathcal{B}$ is a basis for T , then $p \in \bar{A}$ if and only if for all basis elements $B \in \mathcal{B}$ containing p , we have $B \cap A \neq \emptyset$.

Proof.

□

§ 1.7. Limit Points of a Set

Definition 1.7.1. Let (X, T) be a topological space. Let $A \subseteq X$. A point $p \in X$ is a limit point of A if $p \in \bar{A} \setminus \{p\}$. The set of limit points of A is denoted A' .

Example 1.7.1. Consider $\mathbb{R}$ with the standard topology.

- (1) If F is a finite set, then $F' = \emptyset$.
- (2) For any $a < b$, we have $(a, b]' = [a, b]$.
- (3)
- (4) The set containing the limit points of $\{\frac{1}{n} \mid n \in \mathbb{N}\}$ is $\{0\}$.

Example 1.7.2. Let X be a set with the discrete topology. For any $A \subseteq X$, the set containing the limit points of A is empty.

Proposition 1.7.1. *Let (X, T) be a topological space. Let $A \subseteq X$. Then A is closed in X if and only if A contains all of its limit points.*

Proof.

$$\begin{aligned}
 A \text{ is closed in } X &\iff A^c \text{ is open in } X \\
 &\iff (\forall p \in A^c)(\exists U \in T) : p \in U \subseteq A^c \\
 &\iff (\forall p \in A^c)(\exists U \in T) : p \in U \text{ and } U \cap A = \emptyset \\
 &\iff (\forall p \in A^c) : p \notin \overline{A} \\
 &\iff (\forall p \in A^c) : p \notin \overline{A \setminus \{p\}} \\
 &\iff (\forall p \in A^c) : p \notin \overline{A \setminus \{p\}} \\
 &\iff A \text{ contains its limit points.} \quad \square
 \end{aligned}$$

Proposition 1.7.2. *Let (X, T) be a topological space and $A \subseteq X$. Then $\overline{A} = A \cup A'$.*

Proof. Let $p \in \overline{A}$. If $p \in A$, then by Proposition 1.7.1 we are done. Suppose $p \in \overline{A} \setminus A$. Since $p \in \overline{A}$, for all open sets $U \subseteq X$ containing p , we have $U \cap A \neq \emptyset$. But since $p \notin A$, this is equivalent to the statement that for all open sets $U \subseteq X$ containing p , we have $U \cap (A \setminus \{p\}) \neq \emptyset$. Therefore $p \in \overline{A \setminus \{p\}}$. This establishes $\overline{A} \subseteq A \cup A'$.

Now let $p \in A \cup A'$. If $p \in A$, then $p \in \overline{A}$. If $p \in A'$, then $p \in \overline{A}$ by Proposition 1.7.1. Hence $\overline{A} = A \cup A'$. $\square$

§ 1.8. Hausdorff Spaces

Definition 1.8.1. Let (X, T) be a topological space. We say X is *Hausdorff* (or T_2) if for all $x, y \in X$, there exist an open set $U \subseteq X$ containing x and an open set $V \subseteq X$ containing y such that $U \cap V = \emptyset$.

Example 1.8.1. The space $(\mathbb{R}^n, \tau_{\mathbb{R}^n})$ is Hausdorff. For any $x, y \in \mathbb{R}^n$, let $\epsilon := \frac{\|x-y\|}{3}$. Then $B(x, \epsilon) \cap B(y, \epsilon) = \emptyset$.

Example 1.8.2. The space $(X, \mathcal{P}(X))$ is Hausdorff since for any $x, y \in X$, we have $\{x\} \cap \{y\} = \emptyset$.

Example 1.8.3. The trivial topology on a set X is not Hausdorff.

Example 1.8.4. Let $X := \{a, b, c\}$ and $T := \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, \{a, b, c\}\}$. Then (X, T) is not Hausdorff. Indeed, the elements b and c cannot be separated by disjoint open sets, as any open set which contains c also contains b .

Proposition 1.8.1. *In any Hausdorff space, every finite set is closed.*

Proof. Let $p, q \in X$. Then there exists an open set U containing p and an open set V containing q such that $U \cap V = \emptyset$. Now $p \in X \setminus \{q\}$. Hence $p \in U \subseteq X \setminus \{q\}$. Therefore

$X \setminus \{q\}$ is open, establishing that $\{q\}$ is closed. Since any finite set can be written as the finite union of singletons, every finite set in a Hausdorff space is closed. $\square$

Proposition 1.8.2. *Let (X, T) be Hausdorff and $A \subseteq X$. Then a point $p \in X$ is a limit point of A if and only if every open set $U \subseteq X$ which contains p also contains infinitely many points of A .*

Proof. We will first prove the backwards direction. Let $U \subseteq X$ be open and $p \in U$. If U contains infinitely many points of A , then U will contain infinitely many points of $A \setminus \{p\}$. Hence $U \cap (A \setminus \{p\}) \neq \emptyset$. Since U was arbitrary, $p \in \overline{A \setminus \{p\}}$, which means p is a limit point of A .

Now let p be a limit point of A . Suppose towards contradiction there exists an open set $U \subseteq X$ containing p such that $\text{card}(U \cap A) < \infty$. Then we also have $\text{card}(U \cap (A \setminus \{p\})) < \infty$. Suppose $U \cap (A \setminus \{p\}) = \{a_1, \dots, a_n\}$. Since X is Hausdorff, $X \setminus \{a_1, \dots, a_n\}$ is open. In particular, $U \cap (X \setminus \{a_1, \dots, a_n\})$ is open and contains p . But this means:

$$\begin{aligned} (U \cap (X \setminus \{a_1, \dots, a_n\})) \cap (A \setminus \{p\}) &= (U \cap (A \setminus \{p\})) \cap (X \setminus \{a_1, \dots, a_n\}) \\ &= \{a_1, \dots, a_n\} \cap (X \setminus \{a_1, \dots, a_n\}) \\ &= \emptyset. \end{aligned}$$

Since there exists an open set containing p which is disjoint from $A \setminus \{p\}$, $p \notin \overline{A \setminus \{p\}}$. This is a contradiction, as we assumed p to be a limit point of A . $\square$

Definition 1.8.2. In an arbitrary topological space X , a sequence $\{x_n\}_{n=1}^{\infty} \subset X$ *converges* to a point $x \in X$ provided that, for all neighborhoods U of x , there is a positive integer N such that $x_n \in U$ for all $n \geq N$.

Theorem 1.8.1. *If X is a Hausdorff space, then a sequence of points of X converges to at most one point of X .*

Proof. $\square$

Example 1.8.5. Consider the topological space $(\mathbb{N}, T)$, where T is the cofinite topology. Consider the sequence $\{i\}_{i=1}^{\infty} \subset \mathbb{N}$. Let $p \in \mathbb{N}$ be arbitrary. Let U be a neighborhood of p . Since U is open in the cofinite topology, $\text{card}(U^c) < \infty$. Pick the largest $N \in U^c$. Then for all $i \geq N$ we have $i \in U$. Therefore the sequence $\{i\}_{i=1}^{\infty}$ converges to p . Since p was arbitrary, the sequence $\{i\}_{i=1}^{\infty}$ converges to every point in $\mathbb{N}$.

§ 1.9. Continuity

Definition 1.9.1. Let (X, T_X) and (Y, T_Y) be topological spaces. A function $f : X \rightarrow Y$ is continuous if for all $U \in T_Y$, we have $f^{-1}(U) \in T_X$.

Example 1.9.1. For topological spaces $(X, \mathcal{P}(X))$ and $(Y, \mathcal{T})$, any map $f : X \rightarrow Y$ is continuous.

Example 1.9.2. Consider the sets $X := \{a, b, c, d\}$ and $Y := \{1, 2, 3\}$. Let:

$$\begin{aligned} \mathcal{T}_X &:= \{\emptyset, \{c\}, \{a, b\}, \{a, b, c\}, \{c, d\}, \{a, b, c, d\}\}, \\ \mathcal{T}_Y &:= \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{1, 2, 3\}\}. \end{aligned}$$

Then $f : X \rightarrow Y$ given by:

$$\begin{aligned} f(a) &= 1, \\ f(b) &= 1, \\ f(c) &= 2, \\ f(d) &= 2, \end{aligned}$$

is continuous, while $g : X \rightarrow Y$ given by:

$$\begin{aligned} g(a) &= 1, \\ g(b) &= 2, \\ g(c) &= 2, \\ g(d) &= 3, \end{aligned}$$

is not continuous.

Proposition 1.9.1. Let X, Y be topological spaces. Let $f : X \rightarrow Y$ be a map. The following are equivalent:

- (1) f is continuous;
- (2) For all $A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$;
- (3) $f^{-1}(C)$ is closed in X for any C closed in Y .
- (4) For all $x_0 \in X$ and for all $V \in \mathcal{T}_Y$ such that $f(x_0) \in V$, there exists a $U \in \mathcal{T}_X$ such that $x_0 \in U$ and $f(U) \subseteq V$.

Proof.

□

Example 1.9.3. The following is a counter example of (2) from Proposition 1.9.1 when f is not continuous. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) := \chi_{\mathbb{Q}}(x)$. Then $f(\overline{\mathbb{Q}}) = f(\mathbb{R}) = \{0, 1\}$, whilst $\overline{f(\mathbb{Q})} = \overline{\{1\}} = \{1\}$.

§ 1.10. Metric Topologies

Definition 1.10.1. A metric topology is the topology generated by the basis of open balls defined by the given metric.

Definition 1.10.2. Let $(X, \mathcal{T})$ be a topological space. We say X is metrizable if there exists a metric d that induces the given topology on X .

Example 1.10.1. The topological space $(X, \mathcal{P}(X))$ is metrizable.

Proposition 1.10.1. Every metrizable topological space is Hausdorff.

Proof.

□

Example 1.10.2. Consider the three point set $\{a, b, c\}$ with the topology

$$\mathcal{T} := \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, \{a, b, c\}\}.$$

Since this topological space is not Hausdorff, it is not metrizable.

Definition 1.10.3. Let (X, d) be a metric space and $E \subseteq X$. The diameter of E is $\text{diam}(E) := \sup\{d(x, y) \mid x, y \in E\}$. If $\text{diam}(E) < \infty$, we say E is bounded.

Proposition 1.10.2. Let (X, d) be a metric space. Define $\bar{d} : X \times X \rightarrow \mathbb{R}$ by $\bar{d}(x, y) := \min\{d(x, y), 1\}$. Then $\bar{d}$ is a metric and induces the same topology as d .

Theorem 1.10.1. Let X be a topological space. Let $A \subseteq X$. If $\{x_n\}_{n=1}^{\infty} \subset A$ is a sequence which converges to some $x_0 \in X$, then $x_0 \in \bar{A}$. The converse holds if X is metrizable.

Proof. Let U be a neighborhood of x_0 . Find N sufficiently large such that for all $n \geq N$ we have $x_n \in U$. Since $\{x_n\}_{n=1}^{\infty} \subset U$, we have $x_n \in U \cap A$. Since $U \cap A$ is nonempty for every neighborhood of x_0 , this means $x_0 \in \bar{A}$.

Conversely, suppose X is metrizable. There exists a metric d which induces the same topology on X . As $x_0 \in \bar{A}$, we can inductively find a sequence $\{x_n\}_{n=1}^{\infty}$ such that $x_n \in B(x_0, \frac{1}{n}) \cap A$. Hence $\{x_n\}_{n=1}^{\infty} \subset A$ and also converges to x_0 . □

Example 1.10.3. Consider the three point set $X := \{a, b, c\}$ with the topology

$$\mathcal{T} := \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, \{a, b, c\}\}.$$

If $A \subseteq X$ is a neighborhood of b , then $\bar{A} = X$. Hence for each $p \in \bar{A}$, we have a sequence $\{b\}_{n=1}^{\infty}$ converging to p . Since X is not Hausdorff, it is not metrizable, so the converse of Theorem ?? does not hold.

§ 1.11. Quotient Topologies

Definition 1.11.1. Let X, Y be topological spaces and $p : X \rightarrow Y$ a continuous surjection. We say p is a quotient map if U is open in Y if and only if $p^{-1}(U)$ is open in X .

Remark. An equivalent condition is to require that a subset A of Y be closed in Y if and only if $p(A)$ is closed in X . It follows from the equation $p^{-1}(Y \setminus B) = X \setminus p^{-1}(B)$.

Proposition 1.11.1. *An open (or closed) continuous surjection is a quotient map.*

Proof.

□

Remark. Note that continuous surjections need not be quotient maps, and quotient maps need not be open (or closed) maps.

Example 1.11.1. Define $f : (\mathbb{R}, \mathcal{P}(\mathbb{R})) \rightarrow (\mathbb{R}, \tau_{\mathbb{R}})$ by $f(x) = x$. This map is continuous, surjective, but is not a quotient map. Note that $p^{-1}(\{3\})$ is open in $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$ but not in $(\mathbb{R}, \tau_{\mathbb{R}})$.

Example 1.11.2. Let $X := [0, 1] \cup [2, 3] \subseteq \mathbb{R}$ and $Y = [0, 2] \subseteq \mathbb{R}$. Define $p : X \rightarrow Y$ by:

$$p(x) := \begin{cases} x, & x \in [0, 1] \\ x - 1, & x \in [2, 3] \end{cases}.$$

Then p is closed, continuous, and a surjection. Hence it is a quotient map. But $[0, 1]$ is open in X , whilst $p([0, 1]) = [0, 1]$ is not open in Y . Thus p is not an open map.

Proposition 1.11.2. *Let X be a topological space and Y any set. Let $p : X \rightarrow Y$ be a surjection. The set $\{U \subseteq Y \mid p^{-1}(U) \text{ open in } X\}$ is the unique topology on Y which makes p a quotient map.*

Proof.

□

Definition 1.11.2. The topology in Proposition 1.11.2 is called the quotient topology on Y induced by p .

Proposition 1.11.3. *Let (X, τ) be a topological space and Y a set. Let $p : X \rightarrow Y$ be a surjection. The quotient topology on Y induced by p is the finest topology on Y such that p is continuous.*

Proof.

□

Example 1.11.3. Define $p : \mathbb{R} \rightarrow \{a, b, c\}$ by:

$$p(x) := \begin{cases} a, & x > 0 \\ b, & x < 0 \\ c, & x = 0 \end{cases}.$$

Then the set:

$$\{X \subseteq \{a, b, c\} \mid p^{-1}(X) \text{ open in } \mathbb{R}\} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, b, c\}\}$$

is the quotient topology on Y induced by p .

Definition 1.11.3. Let $\sim$ be an equivalence relation on a topological space X . Let $X/\sim$ denote the set of distinct equivalence classes. Let $p : X \rightarrow X/\sim$ be the canonical surjection given by $p(x) = [x]$. Let $T_\sim$ be the quotient topology on $X/\sim$ induced by p . We call the space $(X/\sim, T_\sim)$ the quotient space of X modulo $\sim$.

Example 1.11.4. Consider the set $S := \{(x, y) \mid x \in [0, 1], y \in [0, 1]\}$. Define $\sim$ on S by:

$$\begin{aligned} (0, y) &\sim (1, y) \text{ for all } y \in [0, 1] \\ (x, y) &\sim (x, y) \text{ for all } (x, y) \in (0, 1) \times [0, 1]. \end{aligned}$$

This is an equivalence relation. The space $(S/\sim, T_\sim)$ is homeomorphic to a cylinder in $\mathbb{R}^3$.

Example 1.11.5. Let $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. Define $\sim$ on D by:

$$\begin{aligned} (x, y) &\sim (x, y) \text{ whenever } x^2 + y^2 < 1, \\ (x, y) &\sim (a, b) \text{ whenever } x^2 + y^2 = 1. \end{aligned}$$

This is an equivalence relation. The space $(D/\sim, T_\sim)$ is homeomorphic to a sphere in $\mathbb{R}^3$.

Corollary 1.11.1. Let $g : X \rightarrow Z$ be a surjective continuous map. Let $X^* = \{g^{-1}(\{z\}) \mid z \in Z\}$. Give X^* the quotient topology. Then g induces a bijective continuous map $f : X^* \rightarrow Z$ which is a homeomorphism if and only if g is a quotient map. Moreover, the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{g} & Z \\ p \downarrow & \nearrow f & \\ X^* & & \end{array}$$

Proof.

□

Example 1.11.6. Define $\sim$ on $[0, 1] \times [0, 1]$ by:

$$\begin{aligned} (0, y) &\sim (1, y) \text{ for all } y \in [0, 1] \\ (x, 0) &\sim (x, 1) \text{ for all } x \in [0, 1] \\ (x, y) &\sim (x, y) \text{ for all } (x, y) \in (0, 1) \times (0, 1). \end{aligned}$$

We will prove that $([0, 1] \times [0, 1])/\sim$ is homeomorphic to $S^1 \times S^1$. Define $g : [0, 1] \times [0, 1] \rightarrow S^1 \times S^1$ by $g(s, t) = (\cos 2\pi s, \sin 2\pi s, \cos 2\pi t, \sin 2\pi t)$. This map is continuous and surjective.

Example 1.11.7. Let $p : X \rightarrow Y$ be a quotient map and $A \subseteq X$ a subspace. Then $p|_A : A \rightarrow p(A)$ is not necessarily a quotient map. Indeed, let $X := [0, 1] \cup [2, 3] \subseteq \mathbb{R}$ and $Y := [0, 2] \subseteq \mathbb{R}$. Define $p : X \rightarrow Y$ by:

$$p(x) := \begin{cases} x, & x \in [0, 1] \\ x - 1, & x \in [2, 3] \end{cases}.$$

Since p is a closed map, it is a quotient map. Consider the subset $A := [0, 1] \cup [2, 3] \subseteq X$. We have $(p|_A)^{-1}([1, 2]) = [2, 3]$. Note that $[2, 3]$ is open in A but $[1, 2]$ is not open in Y .

Example 1.11.8. Let $p : A \rightarrow B$ and $q : C \rightarrow D$ be quotient maps. Then the product $p \times q : A \times C \rightarrow B \times D$ is not necessarily a quotient map.

Chapter 2

Connectedness and Compactness

§ 2.1. Connected Spaces

Definition 2.1.1. Let X be a topological space.

- (1) A *separation of X* is a pair U, V of disjoint and nonempty open subsets of X whose union is X . If a separation of X exists, we say X is *disconnected*.
- (2) The space X is *connected* if there does not exist a separation of X .

Proposition 2.1.1. A nonempty topological space X is connected if and only if the only subsets of X that are both open and closed are $\emptyset$ and X .

Proof. Suppose X is connected. Note that $\emptyset, X$ are by definition open and closed. If there were another subset A such that $A^c = A$, then $X = A \cup A^c$ would be a separation. This is a contradiction, hence $\emptyset, X$ are the only open and closed subsets of X .

For the converse, we consider the contrapositive of the above statement. If X is a disconnected topological space, then there exists nonempty open subsets U, V of X such that $U \cap V = \emptyset$ and $U \cup V = X$. Then U and V are closed and open, as $U^c = V$ and $V^c = U$. $\square$

Example 2.1.1. Consider $[0, 1] \cup [2, 3] \subseteq \mathbb{R}$ with the subspace topology. Then $[0, 1] \cup [2, 3]$ is not connected since $U = [0, 1]$ and $V = [2, 3]$ form a separation.

Definition 2.1.2. A topological space is *totally disconnected* if the only connected subsets are singletons.

Example 2.1.2. The rationals are totally disconnected. Let $A \subseteq \mathbb{Q}$ be a subspace with at least two points. Let $p, q \in A$ and suppose without loss of generality $p < q$. Pick an irrational number c between p and q . Define $U := (\infty, c) \cap A$ and $V := (c, \infty) \cap A$. These are open, nonempty, and disjoint subsets of A . But $U \cup V = A \setminus \{c\} = A$. Since A admits a separation, A is disconnected. As this is true for any subspace of $\mathbb{Q}$ with at least two points, we can conclude that $\mathbb{Q}$ is totally disconnected.

Example 2.1.3. Consider the topological space $(X, \mathcal{P}(X))$. If X has at least two points, then X is not connected, as $\{p\}, X \setminus \{p\}$ form a separation.

Lemma 2.1.1. Let Y be a subspace of a topological space X .

- (1) A, B is a separation of Y ;

(2) $Y = A \cup B$, where $A, B \subseteq Y$ are disjoint, nonempty, and $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.

Proof. Let A, B be a separation of Y . Then $Y = A \cup B$, where $A, B \subseteq Y$ are disjoint, nonempty, and open in Y . We only need to show that $\overline{A} \cap B = A \cap \overline{B} = \emptyset$. Suppose towards contradiction we can find some $p \in \overline{A} \cap B$. Since B is open in Y , we can write $B = Y \cap U$, where $U \subseteq X$ is open. Hence $p \in \overline{A} \cap Y \cap U$. Now since $p \in \overline{A}$ and $p \in U$, the set $A \cap U$ is nonempty. Hence we can find some $a \in A \cap U$. But since $A \subseteq Y$, we have $a \in Y \cap U = B$. Thus $a \in A \cap B$, which is a contradiction. By a similar argument, $A \cap \overline{B} = \emptyset$.

Conversely, we must show that A and B are open in Y . Since $\overline{A} \cap B = \emptyset$, note that:

$$\begin{aligned} \overline{A} \cap Y &= \overline{A} \cap (A \cup B) \\ &= (\overline{A} \cap A) \cup (\overline{A} \cap B) \\ &= A \cup \emptyset \\ &= A. \end{aligned}$$

A similar argument will show that $\overline{B} \cap Y = B$. Now since $\overline{A}$ and $\overline{B}$ are closed, by definition this means A and B are closed in Y . Since $A = Y \setminus B$ and $B = Y \setminus A$, they are open in Y as well. $\square$

Example 2.1.4. Consider the following subset of $\mathbb{R}^2$:

$$Y := \{(x, 0) \mid x \in \mathbb{R}\} \cup \{(x, \frac{1}{x}) \mid x > 0\}$$

Note that $\{(x, 0) \mid x \in \mathbb{R}\}$ and $\{(x, \frac{1}{x}) \mid x > 0\}$ are disjoint and closed in $\mathbb{R}^2$. By Lemma 2.1.1, $\{(x, 0) \mid x \in \mathbb{R}\}$ and $\{(x, \frac{1}{x}) \mid x > 0\}$ form a separation of Y .

Lemma 2.1.2. *If C and D form a separation of X , and if Y is a connected subspace of X , then either $Y \subseteq C$ or $Y \subseteq D$.*

Proof. Suppose towards contradiction $Y \cap C \neq \emptyset$ and $Y \cap D \neq \emptyset$. Since C and D are open in X , then $Y \cap C$ and $Y \cap D$ are open in Y . Moreover, their union is Y and their intersection is empty. Therefore, $Y \cap C$ and $Y \cap D$ form a separation of Y , which is a contradiction. $\square$

Theorem 2.1.1. *The union of a collection of connected subspaces of X that have a point in common is connected.*

Proof. Let $\{A_i\}_{i \in I}$ be a collection of connected subspaces of X and let $p \in \bigcap_{i \in I} A_i$. Suppose towards contradiction that U, V is a separation of $\bigcup_{i \in I} A_i$. Then either $p \in U$ or $p \in V$. Without loss of generality, assume $p \in U$. Since each A_i is connected, it must lie entirely in either U or V , and it cannot lie in V because it contains the point $p \in U$. Hence $A_i \subset U$ for all i , which means $\bigcup_{i \in I} A_i \subset U$. But this means that $V = \emptyset$, which is a contradiction. $\square$

Theorem 2.1.2. *Let A be a connected subspace of X . If $A \subseteq B \subseteq \overline{A}$, then B is connected.*

Proof. Suppose towards contradiction $B = C \cup D$ is a separation of B . Then $A \subseteq C$ or $A \subseteq D$; suppose without loss of generality that $A \subseteq C$. Then $\overline{A} \subseteq \overline{C}$. By Lemma 2.1.1, $\overline{C} \cap D = \emptyset$. Therefore $B \cap D = \emptyset$, as $B \subseteq \overline{C}$. This contradicts D being a nonempty subset of B . $\square$

Example 2.1.5. The *Topologist's sine curve* is the set:

$$S := \{(x, \sin \frac{1}{x}) \mid 0 < x \leq 1\} \cup \{(0, y) \mid y \in [-1, 1]\}.$$

Note that the closure of $\{(x, \sin \frac{1}{x}) \mid 0 < x \leq 1\}$ is S . Since $\{(x, \sin \frac{1}{x}) \mid 0 < x \leq 1\}$ is connected, S is connected.

Theorem 2.1.3. *The image of a connected space under a continuous function is connected.*

Proof. Let X be connected and $f : X \rightarrow Y$ continuous. If U and V form a separation of $f(X)$, then $f^{-1}(U)$ and $f^{-1}(V)$ form a separation of X , which is a contradiction. $\square$

Theorem 2.1.4. *A finite Cartesian product of connected spaces is connected*

Proof. We will only prove the theorem for the product of two connected spaces X and Y , as the finite case follows by induction. Choose a point $(a, b) \in X \times Y$. The set $X \times \{b\}$ is connected since it is homeomorphic to X . For each $x_0 \in X$, the set $\{x_0\} \times Y$ is connected since it is homeomorphic to Y . Let $T_{x_0} := (X \times \{b\}) \cup (\{x_0\} \times Y)$. Since $(x_0, b) \in (X \times \{b\}) \cap (\{x_0\} \times Y)$, by Theorem 2.1.1 T_{x_0} is connected. Now $\bigcup_{x_0 \in X} T_{x_0}$ is connected, as $(a, b) \in \bigcap_{x_0 \in X} T_{x_0}$. But $\bigcup_{x_0 \in X} T_{x_0} = X \times Y$, hence $X \times Y$ is connected. $\square$

§ 2.2. Connected Subspaces of $\mathbb{R}$

Theorem 2.2.1. *A set $E \subseteq \mathbb{R}$ is connected if and only if whenever $a < c < b$ with $a, b \in E$, then $c \in E$.*

Proof.

$\square$

Theorem 2.2.2 (Intermediate Value Theorem). *Let X be a connected topological space and $f : X \rightarrow \mathbb{R}$ a continuous map. If $a, b \in X$ and l is a point in $\mathbb{R}$ lying between $f(a)$ and $f(b)$, then there exists $c \in X$ such that $f(c) = l$.*

Proof. Suppose such a c does not exist. Let $A := (-\infty, l) \cap f(X)$ and $B := (l, \infty) \cap f(X)$. Clearly A and B are disjoint, and moreover $A \cup B = f(X) \setminus \{l\} = f(X)$. This means A and B form a separation of $f(X)$. But $f(X)$ is connected as f is continuous—a contradiction. $\square$

§ 2.3. Path Connectedness

Definition 2.3.1. For $x, y \in X$, a path in X from x to y is a continuous function $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x$ and $\gamma(1) = y$. The space X is called path connected if every pair of points can be joined by a path.

Example 2.3.1. Any convex subset of a topological vector space is path-connected via a straight line.

Example 2.3.2. Let K be a finite subset of $\mathbb{R}^n$. Then $\mathbb{R}^n \setminus K$ is path connected. This is not true for $n = 1$.

Proposition 2.3.1. If X is path connected and $f : X \rightarrow Y$ is continuous, then $f(X)$ is path connected.

Proof. Let $y_1, y_2 \in f(X)$. Then there exists $x_1, x_2 \in X$ such that $y_1 = f(x_1)$ and $y_2 = f(x_2)$. Since X is path connected, there exists a continuous map $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x_1$ and $\gamma(1) = x_2$. Since f and γ are continuous, the composition $f \circ \gamma$ is continuous. Hence $f \circ \gamma : [0, 1] \rightarrow Y$ is a continuous map such that $(f \circ \gamma)(0) = f(x_1) = y_1$ and $(f \circ \gamma)(1) = f(x_2) = y_2$. Thus $f(X)$ is path connected. $\square$

§ 2.4. Compactness

Definition 2.4.1. Let X be a topological space. A cover of X is a collection $\{U_i\}_{i \in I} \subset X$ such that $\bigcup_{i \in I} U_i = X$. If U_i is open for all $i \in I$, we say $\{U_i\}_{i \in I}$ is an open cover of X . A subcover of $\{U_i\}_{i \in I} \subset X$ is a sub-collection of $\{U_i\}_{i \in I}$ whose union is X . A finite subcover is a finite subcollection of $\{U_i\}_{i \in I}$ that still covers X .

Definition 2.4.2. A topological space X is compact if every open cover of X admits a finite subcover.

Example 2.4.1. The set $\{(n-1, n+1) \mid n \in \mathbb{N}\}$ is an open cover of $\mathbb{R}$ which does not admit a finite subcover. Hence $\mathbb{R}$ is *not* compact.

Example 2.4.2. The set $\{(\frac{1}{n+1}, \frac{1}{n}) \mid n \in \mathbb{N}\}$ is an open cover of $(0, 1)$ which does not admit a finite subcover.

Example 2.4.3. Finite topological spaces are compact. If $X = \{x_1, \dots, x_n\}$ and $\{U_i\}_{i \in I}$ is an open cover of X , then for each $x_i \in X$ pick $U_{x_i} \in \{U_i\}_{i \in I}$ such that $x_i \in U_{x_i}$. Then $\bigcup_{i=1}^n U_{x_i}$ is a finite subcover.

Example 2.4.4. The topological space $(X, \mathcal{P}(X))$ is not compact. Indeed, $\bigcup_{x \in X} \{x\}$ is an open cover which does not admit a finite subcover.

Proposition 2.4.1. *Let Y be a subspace of a topological space X . Then Y is compact if and only if every collection $\{U_i\}_{i \in I}$ of open sets in X that contains Y has a finite subcollection $\{U_1, \dots, U_n\}$ also containing Y .*

Proof.

□

Proposition 2.4.2. *Every closed subspace of a compact space is compact.*

Proof. Let X be a compact topological space and let $Y \subseteq X$ be closed. Let $\{U_i\}_{i \in I}$ be an open cover of Y . Then $\{U_i\}_{i \in I} \cup Y^c$ is an open cover of X . Since X is compact, we can find a finite subcover $\{U_1, \dots, U_n\} \cup Y^c$. This gives $Y \subseteq \bigcup_{i=1}^n U_i$, which means Y is compact. □

Lemma 2.4.1 (A point and a compact set can be separated). *Let X be a Hausdorff topological space, $K \subseteq X$ a compact subset, and $x_0 \notin K$. Then there exists open sets U and V such that $x_0 \in U$, $K \subseteq V$, and $U \cap V = \emptyset$.*

Proof. Since X is Hausdorff, for every $k \in K$ there exists open sets U_k and V_k such that $x_0 \in U_k$, $y \in V_k$, and $U_k \cap V_k = \emptyset$. Then $K \subseteq \bigcup_{k \in K} V_k$. Since K is compact, there exists a finite subcover; i.e., $K \subseteq \bigcup_{i=1}^n V_{k_i}$. Since each U_{k_i} is a neighborhood of x_0 which is disjoint from V_{k_i} , it must be the case that $(\bigcap_{i=1}^n U_{k_i}) \cap (\bigcup_{i=1}^n V_{k_i}) = \emptyset$. This establishes the claim. □

Proposition 2.4.3. *A compact subset of a Hausdorff space is closed.*

Proof. Let X be Hausdorff and $K \subseteq X$ compact. Let $x_0 \in K^c$. By Lemma ?? there exists open sets U and V such that $x_0 \in U$, $K \subseteq V$, and $U \cap V = \emptyset$. But $U \cap K = \emptyset$ implies $U \subseteq K^c$. Thus $x_0 \in U \subseteq K^c$, and since x_0 is arbitrary this establishes that K is closed. □

Example 2.4.5. Consider the three point set $\{a, b, c\}$ with the topology:

$$\tau := \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, \{a, b, c\}\}.$$

Then $\{b\}$ is compact but not closed.

Proposition 2.4.4. *The continuous image of a compact set is compact.*

Proof. Let $f : X \rightarrow Y$ be continuous. Let $\{U_i\}_{i \in I}$ be a collection of open sets in Y that cover $f(X)$. Then $\bigcup_{i \in I} f^{-1}(U_i)$ is an open cover of X . Since X is compact, there exists a finite subcover $\{f^{-1}(U_1), \dots, f^{-1}(U_n)\}$. Hence $\{U_1, \dots, U_n\}$ is a finite subcover of $f(X)$. □

Proposition 2.4.5. *Let $f : X \rightarrow Y$ be a bijective continuous map. If X is compact and Y is Hausdorff, then f is a homeomorphism.*

Proof. If $C \subseteq X$ is closed, then C is compact. Since $f(C) \subseteq Y$ is compact, $f(C)$ is closed. Thus f is a closed map, which implies it is a homeomorphism. $\square$

Corollary 2.4.1. *If X is compact, Y is Hausdorff, and $p : X \rightarrow Y$ is a continuous surjection, then p is a quotient map.*

Chapter 3

Homotopy Theory

§ 3.1. Homotopy of Functions and Paths

Unless otherwise stated, assume $I := [0, 1]$.

Definition 3.1.1. If f and g are continuous maps from topological spaces X to Y , we say f is *homotopic* to g if there exists a continuous map $F : X \times I \rightarrow Y$ such that $F(x, 0) = f$ and $F(x, 1) = g$. We say F is a *homotopy* between f and g , and write $f \simeq g$. If f is homotopic to a constant map, we say f is *nullhomotopic*.

Example 3.1.1. *** Let X be a topological space and Y a convex subset of a topological vector space. Then for any two functions $f, g : X \rightarrow Y$, we have $f \simeq g$. Indeed, define $F : X \times I \rightarrow Y$ by:

$$F(x, t) := (1 - t)f(x) + tg(x).$$

Since addition and scalar multiplication are continuous, F is continuous. Moreover, since Y is convex, $F(x, t) \in Y$ for all $t \in [0, 1]$. We call F the *straight line homotopy* between f and g .

Example 3.1.2. Let $i : S^1 \rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}$ be the identity map. Define $f : S^1 \rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}$ by $f(x) = (1, 0)$. Then $i \not\simeq f$.

Definition 3.1.2. A *path* is a continuous map $f : I \rightarrow X$. We call $f(0)$ the *initial point* and $f(1)$ the *terminal point*.

Definition 3.1.3. Two paths $f, g : I \rightarrow X$ are *path homotopic* if they have the same initial point x_0 and same terminal point x_1 , and there exists a continuous map $F : I \times I \rightarrow X$ such that:

$$\begin{aligned} F(0, t) &= x_0 & F(1, t) &= x_1 \\ F(s, 0) &= f(s) & F(s, 1) &= g(s) \end{aligned}$$

for all $s, t \in [0, 1]$. We call F a *path homotopy* and write $f \simeq_p g$.

Lemma 3.1.1. Let X and Y be topological spaces. Then $\simeq$ is an equivalence relation on $C(X, Y)$.

Proof.

□

Corollary 3.1.1. *For any set X , then $\simeq_p$ is an equivalence relation on $C(I, X)$.*

Proof. □

Definition 3.1.4. If f is a path in X from x_0 to x_1 and g is a path in X from x_1 to x_2 , we define the product of f and g to be the path $f * g : I \rightarrow X$ given by:

$$(f * g)(s) = \begin{cases} f(2s), & 0 \leq s \leq \frac{1}{2} \\ g(2s - 1), & \frac{1}{2} \leq s \leq 1 \end{cases}.$$

Theorem 3.1.1. *The operation $*$ is well-defined on $C(I, X)/\simeq_p$, defined by the equation $[f] * [g] = [f * g]$. Moreover, the operation $*$ satisfies the following properties:*

- (a) *If $[f] * ([g] * [h])$ is defined, so is $([f] * [g]) * [h]$, and they are equal (Associativity).*
- (b) *Given $x_0 \in X$, let e_{x_0} denote the constant path $e_{x_0} : I \rightarrow X$ given by $e_{x_0}(s) = x_0$. If f is a path in X for x_0 to x_1 , then $[f] * [e_{x_1}] = [f]$ (Identities).*
- (c) *Given a path $f : I \rightarrow X$ from x_0 to x_1 , define $\bar{f} : I \rightarrow X$ by $\bar{f}(s) = f(1-s)$. Then $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_{x_1}]$ (Inverses).*

Proof. □

Remark. Theorem 3.1.1 establishes that $C(I, X)/\simeq_p$ is a *groupoid*. Here $*$ is not a binary operation as it is not necessarily defined for all pairs of elements in $C(I, X)/\simeq_p$. Furthermore, the identity element is not unique for all path-homotopy equivalences.

§ 3.2. The Fundamental Group

Definition 3.2.1. Let X be a topological space. Fix a point $x_0 \in X$. A path in X that begins and ends at x_0 is called a loop in X based at x_0 . The set of all path homotopies of loops based at x_0 is denoted $\Omega(x_0) := \{\alpha \mid \alpha \text{ is a loop in } X \text{ based at } x_0\}$.

Theorem 3.2.1. *The operation $*$ is a binary operation on $\Omega(x_0)/\simeq_p$, and the pair $(\Omega(x_0)/\simeq_p, *)$ is a group.*

Proof. This follows from Theorem 3.1.1. Clearly $*$ is well-defined for any two elements in $\Omega(x_0)/\simeq_p$, and $[e_{x_0}]$ is the unique identity element. □

Definition 3.2.2. The quotient $\Omega(x_0)/\simeq_p$ is called the fundamental group based at x_0 . We denote it by $\pi_1(X, x_0)$.

Example 3.2.1. If Y is a convex subset of a topological vector space and $x_0 \in Y$, then $\pi_1(Y, x_0) = \{[e_{x_0}]\}$. Given a loop α based at x_0 , then we can employ the straight-line homotopy to show that $\alpha \cong e_{x_0}$.

Theorem 3.2.2. Suppose X is a path-connected topological space. Let $p, q \in X$ and let $\gamma : I \rightarrow X$ be a path from p to q . Then the map $\widehat{\gamma} : \pi_1(X, p) \rightarrow \pi_1(X, q)$ defined by $\widehat{\gamma}([\alpha]) = [\bar{\gamma} * \alpha * \gamma]$ is an isomorphism whose inverse is $\widehat{\bar{\gamma}}$.

Proof. For any $[\alpha], [\beta] \in \pi_1(X, p)$, we have:

$$\begin{aligned} \widehat{\gamma}([\alpha] * [\beta]) &= \widehat{\gamma}([\alpha * \beta]) \\ &= [\bar{\gamma} * (a * b) * \gamma] \\ &= [\bar{\gamma} * a * e_p * b * \gamma] \\ &= [(\bar{\gamma} * a * \gamma) * (\bar{\gamma} * b * \gamma)] \\ &= [\bar{\gamma} * a * \gamma] * [\bar{\gamma} * b * \gamma] \\ &= \widehat{\gamma}([a]) * \widehat{\gamma}([b]). \end{aligned}$$

Hence $\widehat{\gamma}$ is a group homomorphism. Similarly, $\widehat{\bar{\gamma}} : \pi_1(X, q) \rightarrow \pi_1(X, p)$ defined by $\widehat{\bar{\gamma}}([\alpha]) = [\gamma * \alpha * \bar{\gamma}]$ is a homomorphism, and is the inverse of $\widehat{\gamma}$. Thus $\widehat{\gamma}$ is an isomorphism. $\square$

Remark. If X is path connected, by the previous theorem $\pi_1(X, x_0)$ is uniquely determined up to isomorphism. In this case, we drop the x_0 in the notation and write $\pi_1(X)$.

Remark. The isomorphism $\widehat{\gamma}$ depends on γ . The isomorphism is independent of the path γ if and only if $\pi_1(X)$ is abelian (Exercise).

Definition 3.2.3. A topological space is simply connected if it is path connected and its fundamental group is trivial.

Example 3.2.2. Any convex subset of a topological vector space is simply connected.

Lemma 3.2.1. Let $f : X \rightarrow Y$ be continuous. If $\alpha : I \rightarrow X$ is a loop in X based at x_0 , then $f \circ \alpha : I \rightarrow Y$ is a loop in Y based at $f(x_0)$.

Proof.

$\square$

Lemma 3.2.2. Let $f : X \rightarrow Y$ be continuous. If $\alpha \simeq_p \beta$, then $f \circ \alpha \simeq_p f \circ \beta$.

Proof. If F is a path homotopy between α and β , then $f \circ F$ is a path homotopy between $f \circ \alpha$ and $f \circ \beta$. $\square$

Theorem 3.2.3. A continuous function $f : X \rightarrow Y$ induces a group homomorphism $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$ defined by $f_*([\alpha]) = [f \circ \alpha]$.

Proof.

□

Remark. Note that if f is one-to-one, then f_* is not necessarily one-to-one. Similarly, if f is onto, f_* is not necessarily onto.

Example 3.2.3. Let $\iota : S^1 \rightarrow \mathbb{R}^2$ be the inclusion map. This map is clearly one-to-one. However, since $\pi_1(S^1) \cong \mathbb{Z}$ and $\pi_1(\mathbb{R}^2)$ is trivial, the induced group homomorphism $\iota_* : \pi_1(S^1) \rightarrow \pi_1(\mathbb{R}^2)$ cannot be one-to-one.

Example 3.2.4. Define $f : \mathbb{R} \rightarrow S^1$ by $t \mapsto e^{it2\pi}$. This map is clearly onto. However, $f_* : \pi_1(\mathbb{R}) \rightarrow \pi_1(S^1)$ cannot be onto.

Proposition 3.2.1.